Forecasting and Macroeconomic Analytics

ECON 522

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Time Series Introduction

Time Series Introduction

- A time series $y_t \in \mathbb{R}^m$ is a process observed in sequence over time: $t = 1, \ldots, n$. Note the subscript t. The time series is univariate if $m = 1$ and multivariate if $m > 1$.
- This Module is primarily focused on univariate time series models.
- Most economic time series are recorded at discrete intervals such as annual, quarterly, monthly, weekly, or daily. The number of observations s per year is called the frequency.
- Because of the sequential nature of time series, we expect that observations close in calender time, e.g. y_t and its lagged value y_{t-1}, will be dependent.
- This type of dependence structure requires a different distributional theory than for cross-sectional since we cannot divide the sample into independent groups.
- Many of the issues which distinguish time series from cross-section econometrics concern the modeling of these dependence relationships.

Introduction

- Recall that one of the most important component of Times Series data was the cyclical component.
- Which assumptions do we need?
- What kind of model could we fit?
- How do we assure the model is correctly specified? What does miss specification means in this context where causality is not the goal?
- How can we consistently and efficiently estimate these models?
- Testing $\rightarrow$ What kind of asymptotic do we need?
- How can these models be used to forecast and which properties these forecasts will have?

Differences and Growth Rates

- It is common to transform many series by taking logarithms, differences and/or growth rates. This is done for a number of different reasons, but the most credible reason is that this is the suitable variable for the desired analysis.
- Many aggregate series are transformed by taking natural logarithms. This flattens the apparent exponential growth.
- The first difference of a series y_t is $\Delta y_t = y_t - y_{t-1}$.
- The second difference is $\Delta^2 y_t = \Delta y_t - \Delta y_{t-1}$. (acceleration)
- Higher-order differences can be defined similarly but are not used in practice.
- The annual, or year-on-year, change of a series y_t with frequency s is $\Delta_s y_t = y_t - y_{t-s}$
- The one-period growth rate is the percentage change from period $t-1$ to period t :

$$q_t = 100\left(\frac{\Delta y_t}{y_{t-1}}\right) = 100\left(\frac{y_t}{y_{t-1}} - 1\right) \approx \log(y_t) - \log(y_{t-1})$$

	y_t	y_{t-1}
$t=1$	4.3	—
2	5.6	4.3
3	7.2	5.6
4	6.1	7.2
5	8	6.1

* you lose an observation when you lag

The Lag Operator

Lag operator Denoted L .

- $Ly_t = y_{t-1}$.
- The lag operator can be raised to powers, e.g. $L^2 y_t = y_{t-2}$.
- We can also form polynomials of it

$$a(L) = a_0 + a_1 L + a_2 L^2 + \ldots + a_p L^p$$
$$a(L)y_t = a_0 y_t + a_1 y_{t-1} + a_2 y_{t-2} + \ldots + a_p y_{t-p}$$

- Lag polynomials can be multiplied. Multiplication is commutative, $a(L)b(L) = b(L)a(L)$.
- Some lag polynomials can be inverted. We define $(1-\rho L)^{-1}$ by the following equality: $(1-\rho L)(1-\rho L)^{-1} \equiv 1$
- If $|\rho| < 1$, then $(1-\rho L)^{-1} = \sum_{i=0}^{\infty} \rho^i L^i$.
- $(1-L) = \Delta$ such that $y_t - y_{t-1} = (1-L)y_t = \Delta y_t$. First difference
- $(1-L)^d = \Delta^d$ such that $y_t - y_{t-d} = (1-L)^d y_t = \Delta^d y_t$. d^{th} difference

Notation

$$L y_t = y_{t-1}$$

$$L^2 y_t = y_{t-2}$$

$$\Delta y_t = y_t - y_{t-1} =$$

$$y_t - L y_t = (1-L) y_t$$

$$y_t - 3 y_{t-1} + 2 y_{t-2}$$

$$(1 - 3L + 2L^2) y_t = a(L) y_t$$

Definitions - WN and MDS

These are the simplest form of Time Series:

Definition 1- White Noise

White noise is a sequence of random variables $\{e_t\}$ such that $E(e_t) = 0, E(e_t e_s) = 0, E(e_t^2) = \sigma^2$

Note: Often we consider independent WN (uncorrelated vs independent) or Gaussian White Noises. Question: Is a WN process predictable?

Definition 2- Martingale Difference Sequence

Let $\{Y_t\}_{t=1}^{\infty}$ denote a sequence of random scalars with $E(Y_t) = 0 \ \forall t$. Let Ω_t denote the information available at time t (including current and lagged values). If $E(Y_t \mid \Omega_t) = 0$ for $t = 2, 3$, then Y_t is said to be martingale difference sequence with respect to Ω_t.

Note: This implies $E(Y_t \mid Y_{t-1}, Y_{t-2}, ..) = 0$. This condition is stronger than serially uncorrelated (only linear functions) but it less than independent as we allow the possibility that $E\left(Y_t^2 \mid Y_{t-1}, ..\right) \neq 0$.

W.N. ℓ_t

$E(\ell_t) = 0$ ← mean of wn is zero

$E(\ell_t^2) = \sigma^2$

* white noise is serially uncorrelated by definition

cov $(\ell_t \ell_s)$

$\underbrace{E(\ell_t \ell_s)}_{\text{Autocovariance}} = 0$

* $Cov(XY) = E(XY) - E(\cancel{X})^{0} E(\cancel{Y})^{0}$

• if x an y have mean zero, then

$cov(yx) = E(xy)$

Why white noise? completely random, rsi

Corr = 0 imply independence? No

* when 2 things are independent, they have nothing to do w eachother

correlation only has to do with a linear relationship.

This situation is true only when we have normality

y_t

Time Series

Suppose I observe $\{y_1, y_2, \ldots, y_T\}$. This sample is a set of T numbers that is only one possible outcome of the stochastic process that generated the data. Each y_t is the realization of the a random variable Y_t that has some density $f_{Y_t}(y_t)$ With unconditional mean

$$E(Y_t) = \int_{-\infty}^{\infty} y_t f_{Y_t}(y_t)\, dy_t = \mu$$

And variance

$$E(Y_t - \mu_t)^2 = \int_{-\infty}^{\infty} (y_t - \mu_t)^2 f_{Y_t}(y_t)\, dy_t = \sigma^2$$

$y_t \longrightarrow$ density function $f(y_t)$

discrete var: $E(X) = \sum x \cdot p(x)$

continuous : $E(X) =$

Autocovariance Function

Each series is a realization of a random variable with a certain mean and variance. T

Definition - Autocovariance

The covariance between observations at different times is called the auto-covariance function:

$$\gamma(k) = \text{COV}\left(y_t, y_{t-k}\right)$$

Properties of the autocovariance function:

1. $\gamma(0) = \text{VAR}\left(y_t\right) = \sigma^2$: i.e variance is constant and does not depend on t.
2. $\gamma(s) = \gamma(-s)$

$$Cov(y_t \cdot y_{t-k}) = \gamma($$

$\gamma(0)$

$\gamma(1)$

$\gamma(2)$

$\gamma(3)$ → how is gdp today correlated w gdp 3 quarters ago, etc

Autocorrelation Function (ACF)

Often correlation is more informative than covariance so people often look at the autocorrelation function:

Definition - Autocorrelation

$$\rho(s) = \frac{\gamma(s)}{\gamma(0)} = \frac{E\left((y_t - \mu)(y_{t-s} - \mu)\right)}{E\left((y_t - \mu)^2\right)}$$

Properties of the autocorrelation function (ACF):

1. As any other correlation $0 \leq \rho(s) \leq 1$
2. $\rho(0) = 1$

Estimation of ACF

ACF can be easily estimated using the sample equivalent. This is called the sample autocorrelation function or **correlogram**:

$$\hat{\rho}(s) = \frac{\frac{1}{T}\sum_{t=s+1}^{T}\left[\left(y_t - \bar{y}\right)\left(y_{t-\tau} - \bar{y}\right)\right]}{\frac{1}{T}\sum_{t=1}^{T}\left(y_t - \bar{y}\right)^2} = \frac{\sum_{t=s+1}^{T}\left[\left(y_t - \bar{y}\right)\left(y_{t-\tau} - \bar{y}\right)\right]}{\sum_{t=1}^{T}\left(y_t - \bar{y}\right)^2}.$$

In large samples:

$$\hat{\rho}(s) \sim N\left(0, \frac{1}{T}\right)$$

So we can test for significance at different horizons (most packages will give you confidence intervals). There are also bootstrap version of C.I. To test the joint significance of all the autocorrelations we can use the Box-Pierce Q-Statistic

$$Q_{BP} = T\sum_{s=1}^{m}\hat{\rho}^2(s),$$

$y_t = c_t$ $\quad B y_{t-3} + \varepsilon_t$

ACF

$$\frac{\mathrm{cov}(x, y)}{\mathrm{var}(x)}$$

$P(3) = B = \mathrm{cov}(y,$

$$y_t = C + B_1 y_{t-1} + B_2 y_{t-2} + B_3 y_{t-3} + \varepsilon$$

PACF

Two different types of correlation

Partial Autocorrelation (PACF)

Definition - Partial Autocorrelation

Partial Autocorrelation (PACF) $p(s)$ is the coefficient of y_{t-s} in a population regression of y_t on $y_{t-1}, y_{t-2}, \ldots, y_{t-s}$.

(this is called an autogression since it is a regression of y_t on itself).

- $p(s)$ measures the autocorrelation after controlling for the effect of $y_{t-1}, y_{t-2}, \ldots, y_{t-s}$.
- Still have that $p(0) = 1$ and. $0 \leq p(s) \leq 1$
- The pattern of the autocorrelation and the partial autocorrelation depend on the series.
- In fact we can predict the model for the time series by looking at patterns in ACF and PACF.

Stationarity

We need what we have observed to be stable, in some sense, so that we can make inferences and statements about the future.

Definition - Strong Stationarity

A process, $\{y_t\}$, is strictly (strongly) stationary if for each k, t and n, the distribution of $\{y_t, \ldots, y_{t+k}\}$ is the same as the distribution of $\{y_{t+n}, \ldots, y_{t+k+n}\}$

If y_t is i.i.d., then it is strictly stationary

Definition - Weak Stationarity

$\{y_t\}$, is 2nd order stationary(or weakly stationary, or simply stationary) if $E(y_t) = \mu$ and $\text{cov}\,(y_t, y_{t-s}) = \gamma_s$ do not depend on t

Note that this also implies that $VAR(y_t) = \sigma^2$ is a constant. γ_s will still depend on the distance between observations, s.

Why Stationarity Matters:

The intuition: We need the data-generating process to be **stable over time** so that patterns we observe in the past will help us predict the future. If the process keeps changing, past data becomes less useful.

Definition 1 - Strong (Strict) Stationarity:

What it says:
The joint distribution of any set of observations $\{y_t, \ldots, y_{t+k}\}$ is **identical** to the distribution of $\{y_{t+n}, \ldots, y_{t+k+n}\}$

In plain English:

- No matter when you look at the series, the probabilistic behavior is the same
- If you shift the time window, the statistical properties don't change
- **This is a very strong requirement** - the entire distribution must be unchanged

Example:

- If y_t is i.i.d. (independent and identically distributed), it's strictly stationary
- Like flipping the same fair coin forever - the process never changes

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Definition 2 - Weak (Covariance) Stationarity:

This is the one we usually work with in practice!

Requirements:

1. **$E(y_t) = \mu$**: The mean is constant over time (doesn't depend on t)
1. **$cov(y_t, y_{t-s}) = \gamma_s$**: The covariance between observations depends **only on the distance (s) between them**, not on the specific time (t)

What this implies:

- **$VAR(y_t) = \sigma^2$** is constant (note at the bottom of slide)
- **γ_s depends on s but not on t**: The correlation structure is time-invariant

Strong Stationarity	**Weak Stationarity**
Entire distribution unchanged	Only mean and covariance structure unchanged
Very restrictive	More practical
Includes all moments	Only needs first two moments (mean, variance)

Practical Interpretation:

Weak stationarity means:

- The series fluctuates around a **constant mean**
- The **variance doesn't change** over time (no increasing/decreasing volatility)
- The correlation between observations **s periods apart** is the same regardless of when you measure it

Non-stationary examples:

- Stock prices (tend to grow/decline over time - changing mean)
- GDP (tends to grow - has a trend)
- Seasonal data without adjustment (variance changes by season)

Question: Does the distinction between "the covariance depends on s but not t" make sense? That's the key to weak stationarity - it's about the **lag distance**, not the **calendar time**.

Examples of Non-stationary

Example (Break)

$$y_t = \begin{cases} \beta + e_t & t \leq k \\ \beta + \lambda + e_t & t > k \end{cases}$$

Example (Random Walk)

$$y_t = y_{t-1} + e_t$$

One can notice that

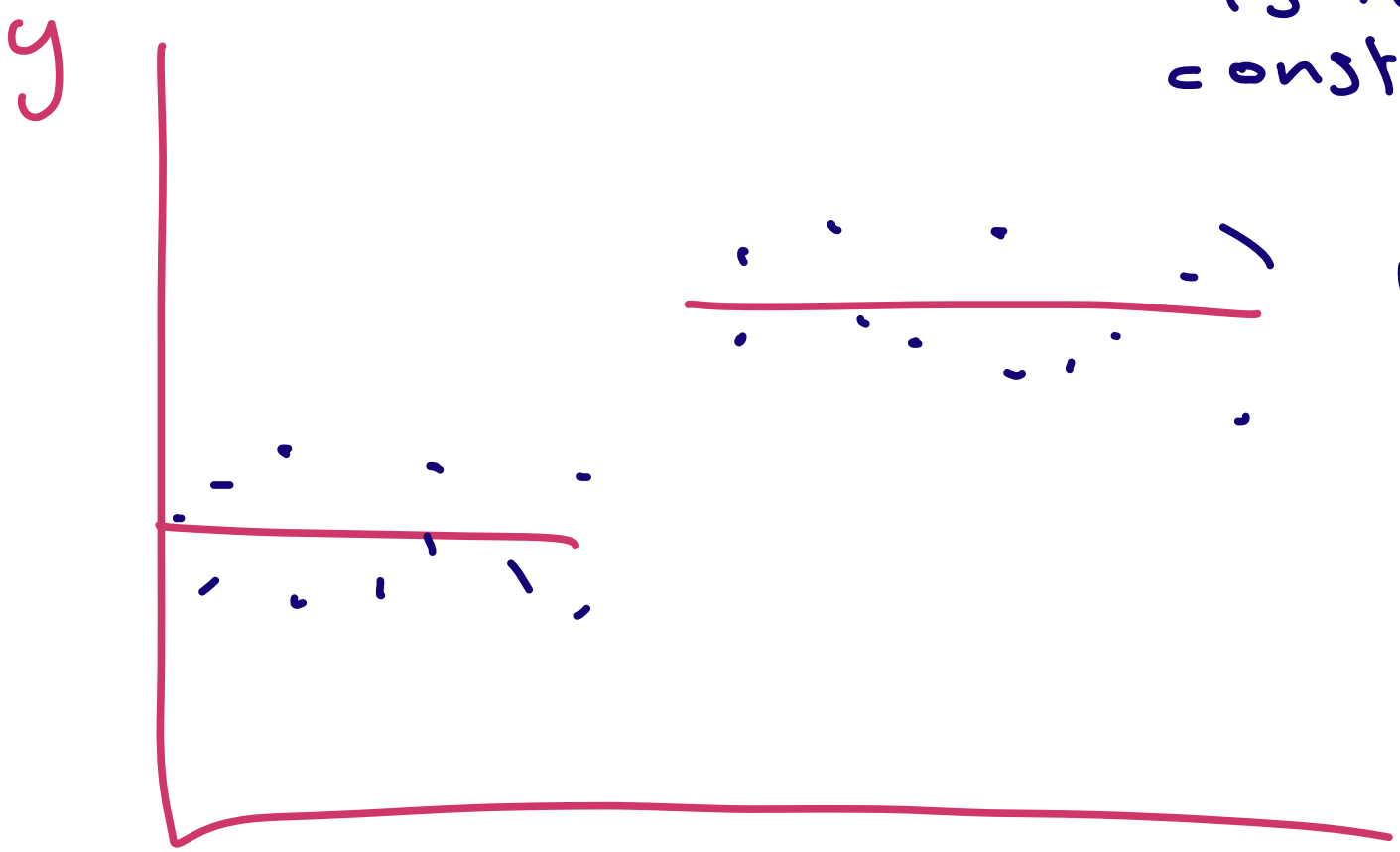
$$\text{Var}(y_t) = \text{Var}(y_{t-1}) + \text{Var}(e_t) > \text{Var}(y_{t-1})$$

$$E(y_t) = E(B + e_t)$$
$$= E(B) + \underbrace{E(e_t)}_{\to 0}$$

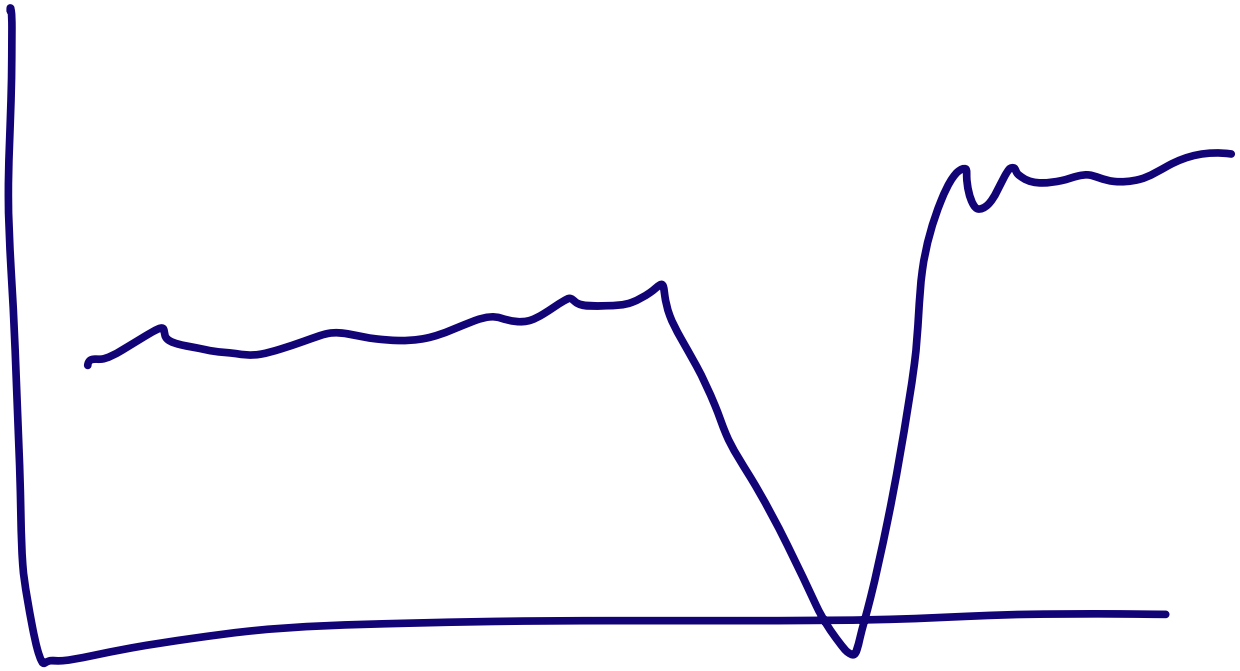
$t \leq k \quad = B$

$t > k$

$E(y_t) - Bt + \lambda$



mean is not constant, so it's not stationary



* difference between outlier and structural break

Ergotic Theorem

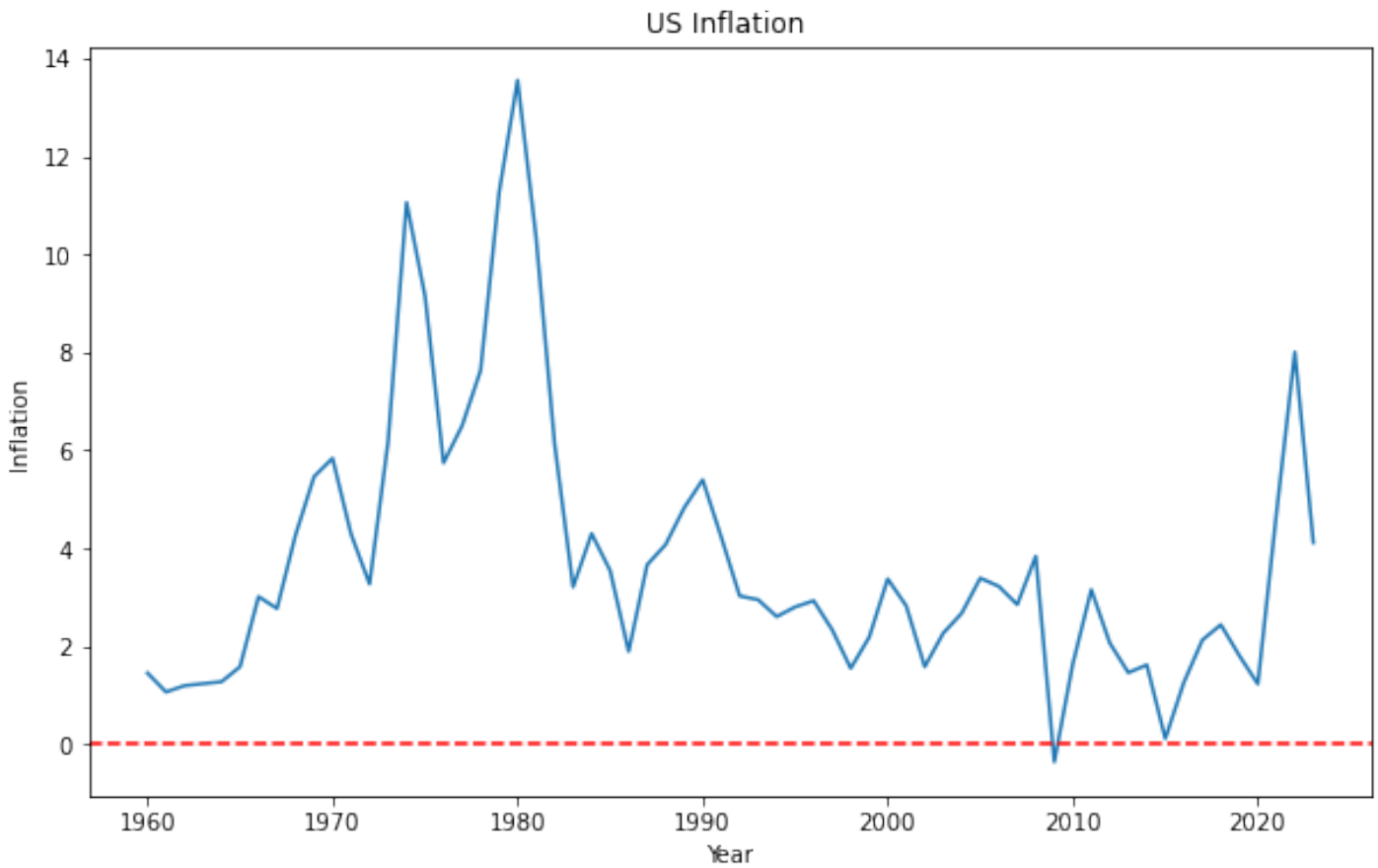
Ergotic Theorem

A covariance stationary process is said to be **ergodic** for the mean if $\bar{y} = \frac{1}{T}\sum_{t=1}^{T} y_t$ converges in probability to $E(Y_t)$ as $T \to \infty$

- A sufficient condition for is that the γ_j autocovariance goes to zero sufficiently fast as j becomes large or that $\sum_{j=0}^{\infty} |\gamma_j| < \infty$
- For many applications, stationarity and ergodicity turn out to be the same requirements, but in theory they are different.
- The implications is that the sample can be used to estimate the mean, variance and covariance (same for the variance = ergodicity for the second moment).

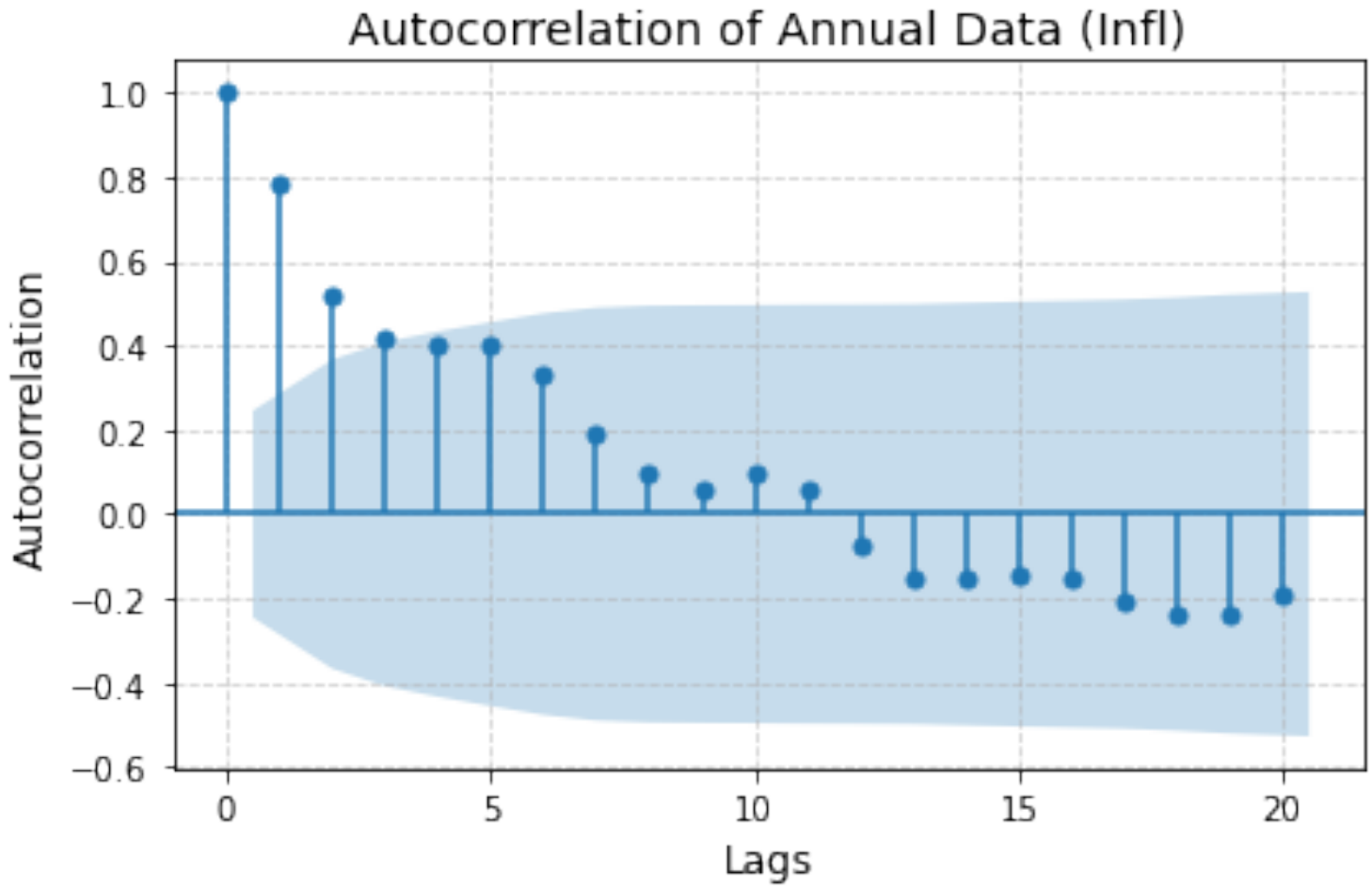
US Inflation Example

Annual Inflation Rate



US Inflation Example - ACF

Inflation Rate is very persistent. Interpret



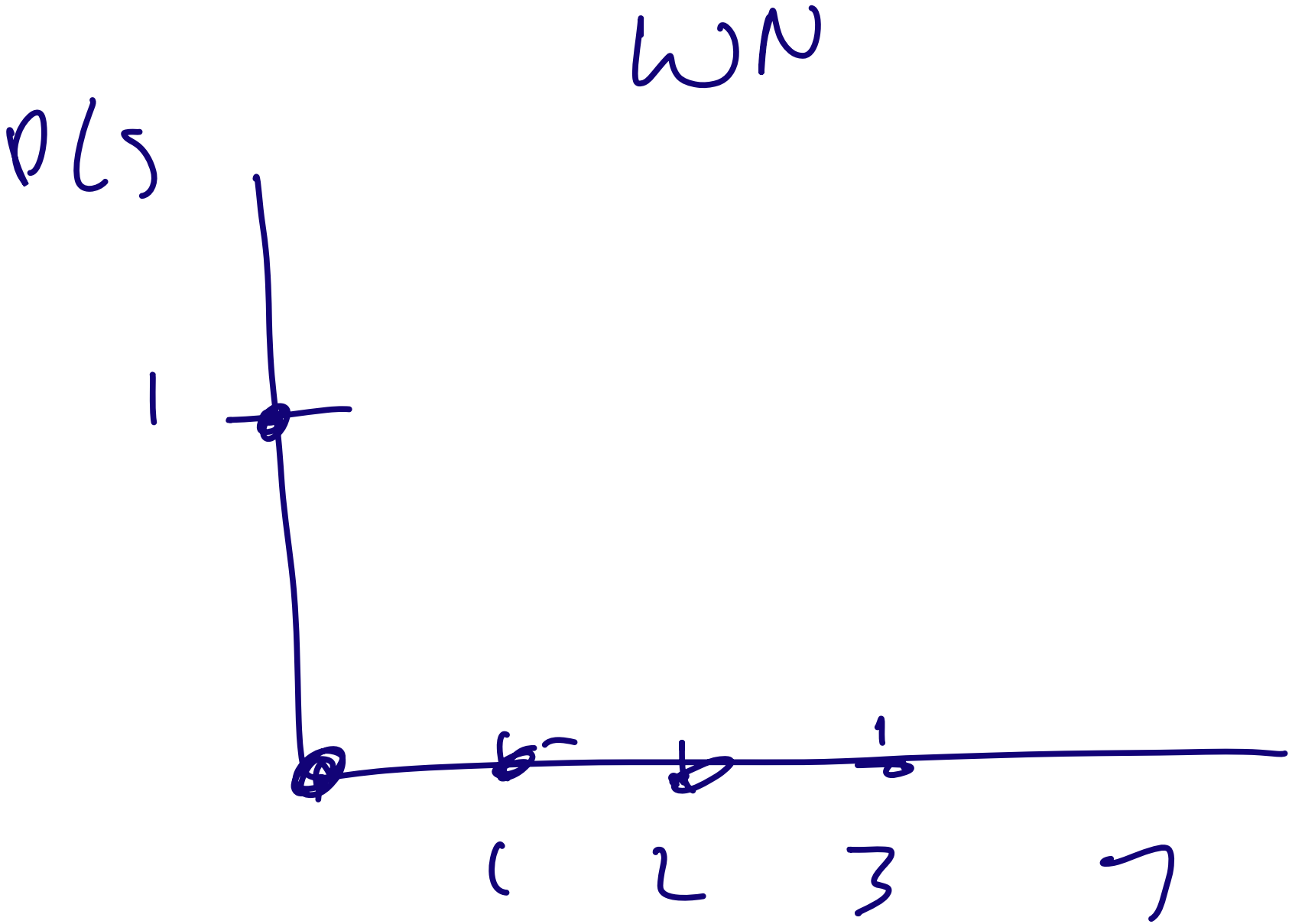
* inflation today has a correlation w/ inflation 1 month ago by 0.8

Models for Cycles

There are various way to model the cycles: there are possible models that would create autocorrelation:

- AR (Autoregressive): errors are function of past error terms.
- MA (Moving average): errors are functions of past other shocks.
- ARMA (Autoregressive Moving Average): a combination of AR and MA.
- White Noise: errors are NOT correlated with the past

We will study each of these models briefly and learn to recognize them.


WN
P(S)
1
1
2
3
7

White Noise

Recall our definition for a White Noise. WN processes look very "random" with no pattern:

$$. \quad \varepsilon_t \sim WN\left(0, \sigma^2\right)$$

1. The mean of WN is zero $E\left(\varepsilon_t\right) = 0$
2. By construction the variance is $\sigma^2 = \gamma(0)$
3. Since WN are serially uncorrelated the autocovariance is zero at any lag:

$$\gamma(s) = \begin{cases} \sigma^2 & s = 0 \\ 0 & s \neq 0 \end{cases}$$

4. The autocorrelation will also be zero at any lag:

$$\rho(s) = \frac{\gamma(s)}{\gamma(0)} = \begin{cases} 1 & s = 0 \\ 0 & s \neq 0 \end{cases}$$

White Noise con't

- ACF will look 1 and 0 (estimation variability but should be not significantly different than zero for any $s > 1$
- To compute the PACF I would have to regress ε_t on its lags. Given that a W.N. is serially uncorrelated all the coefficients on any lag will be zero. (the past does not influence the present).The graph of the PACF will look identical to that of ACF.

- What is your best forecast of a WN? 0
- White noise processes are important because other models (AR and MA) are often function of W.N.
- If you have a good model your residuals or your forecast errors should be WN. Why?

Wold Representation Theorem

Wold Representation Theorem

Any zero mean covariance-stationary process can be represented in the form of

$$Y_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} + k_t$$

Where k_t is the deterministic part, $\psi_0 = 1, and \sum_{j=0}^{\infty} \psi_j^2 < \infty$ ε_t is a white noise which represents the error in forecasting Y on the basis of a linear function of past values.

$$\varepsilon_t = Y_t - \hat{E}\left(Y_t \mid Y_{t-1}, Y_{t-2}, \ldots\right)$$

In theory the theorem means that we need to estimate an infinite number of parameters. With sufficient smoothness on the data we will be able to approximate a MA or infinite order with one of finite order.

Recall that any stationary and invertible $ARMA(p, q)$ can be written as an infinite MA model.

ARMA Models

- The Wold theorem tell us that a stationary TS can be represented by an infinite MA models.
- Is that useful? Can we estimate an infinite number of parameters?
- Can we find a more parimonious model to approximate this representation?
- The answer is yes, as we will see. AR, MA and ARMA models
- We will tackle this next.
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